Neap

VCE Specialist Mathematics Units 3&4

Question and Answer Booklet

2025 Trial Examination 1

Reading time: 15 minutes

Writing time: 1 hour

Student's Name: _____

Teacher's Name: _____

Materials supplied

- Question and Answer Booklet of 15 pages
- Formula Sheet

Instruction

- Write **your name** and your **teacher's name** in the spaces above on this page.

Students are **not** permitted to bring any technology (calculators or software) or notes of any kind into the examination room.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

10 questions (40 marks) 2–15

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Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English.
- Unless otherwise specified, an **exact** answer is required for each question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, diagrams in this booklet are **not** drawn to scale.
- Take the **acceleration due to gravity** to have a magnitude g m s^{-2}, where $g = 9.8$.

Question 1 (4 marks)

An object is moving along a straight line. When the object is x m from the fixed point O,

its acceleration is given by

$$a = \frac{kx^2}{\sqrt{v}},$$

where v is the object's velocity in m s^{-1} and k is a real constant.

When $x = 1$ m, the object's acceleration is 3 m s^{-2}, and its velocity is 4 m s^{-1}.

Find the object's velocity when $x = 2$ m. Give your answer in the form $p^{\frac{m}{n}}$, where $p, m, n \in Z^+$.

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Question 2 (6 marks)

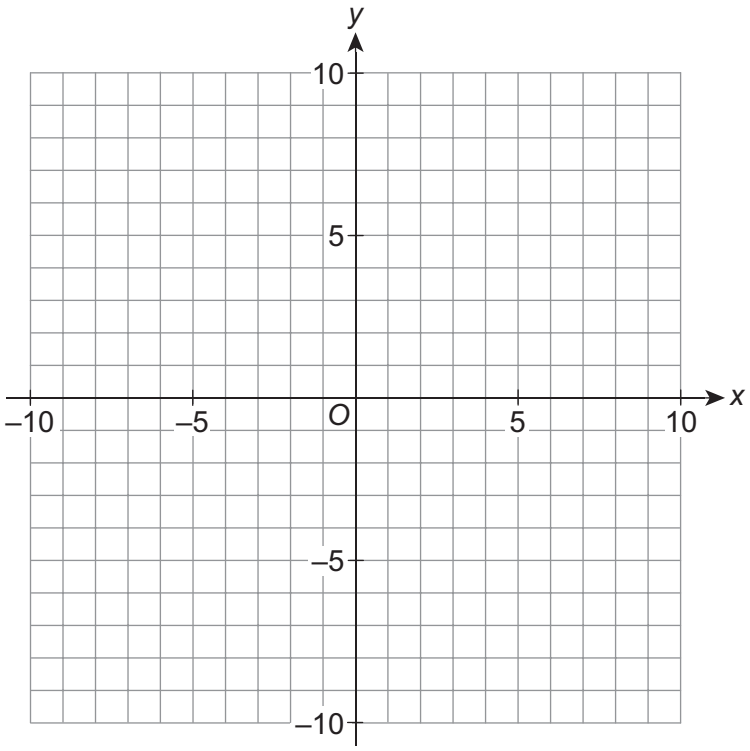
Consider the function f with rule $f(x) = \dfrac{qx^2}{x^2 + 5x - 6}$, where q is a positive real constant.

a. Find the equations of all asymptotes of f. Give your answers in terms of q where required. 2 marks

b. Find the x-values of the stationary points of f. 1 mark

c. Let $q = -1$.

Sketch the graph of f for $x < 1$ on the axes below. Label any stationary points with their coordinates and the asymptotes with their equations. 3 marks



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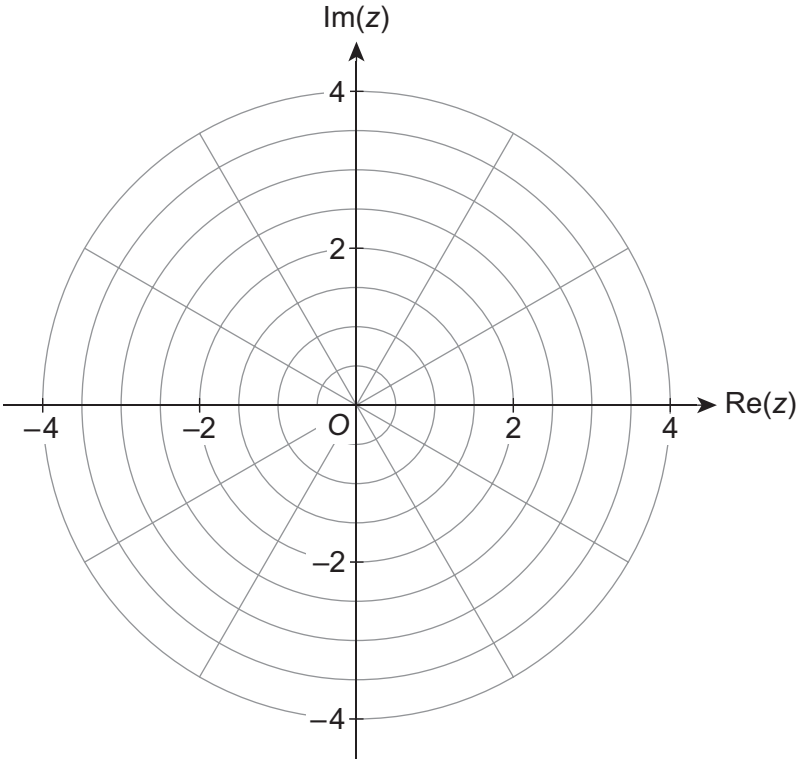
Question 3 (5 marks)
Consider the equation $z^3 + 64i = 0$, where $z \in C$.

a. Solve the equation for z. Give your answers in cartesian form. 2 marks

b. Plot the solutions of the equation on the Argand diagram below. 1 mark



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c. The solutions to the equation $z^3 + 64i = 0$ lie on the circumference of a circle. Two of the solutions are in the third and fourth quadrants, respectively. These solutions lie on a line that divides the circle into two segments.

Find the area of the larger segment. 2 marks

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Question 4 (5 marks)
Consider the points $A(-1, 0, 3)$, $B(0, 2, -3)$ and $D(-3, 1, 0)$.

a. Find the coordinates of C such that $ABCD$ is a parallelogram. 1 mark

b. Points A, B, C and D lie on the plane Π_1.
The plane Π_2 passes through the origin and is parallel to Π_1.
Find the Cartesian equation of Π_2. Give your answer in the form
$px + qy + rz = 0$, where $p, q, r \in R$. 2 marks

c. Find the sine of the angle between the line segment OD and Π_2. 2 marks

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Question 5 (4 marks)

A furniture company makes stools. Each stool consists of one seat, four legs and two horizontal supports between the legs.

The masses of each stool component are normally distributed, such that:

- the seat has a mean mass of 800 g and a standard deviation of 3g
- the leg has a mean mass of 1200 g and a standard deviation of 4 g
- the support has a mean mass of 300 g and a standard deviation of 2 g.

a. Find the mean and standard deviation of the total mass of a stool. 2 marks

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b. Let $\Pr(-2 < Z < 2) = 0.95$ and $\Pr(-1 < Z < 1) = 0.68$.

For a random sample of 36 stools, find the probability that the mean mass of a stool is between 6197 g and 6201.5 g. Give your answer correct to three decimal places. 2 marks

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Question 6 (3 marks)

Use mathematical induction to prove that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$, where n is a natural number.

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Question 7 (3 marks)

Find and simplify $\frac{dy}{dx}$ for the relation $x\sin\left(e^{y}\right)+x^{3}\log_{e}\left(\frac{1}{y}\right)=0$.

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Question 8 (3 marks)

A parametric curve is defined by $x = 6t$ and $y = t^2 - \frac{9}{2}\log_e(t)$, where $t > 0$.

The curve is rotated about the y-axis between $t = 2$ and $t = 4$ to form a solid of revolution.

Find the surface area of the solid of revolution.

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Question 9 (4 marks)

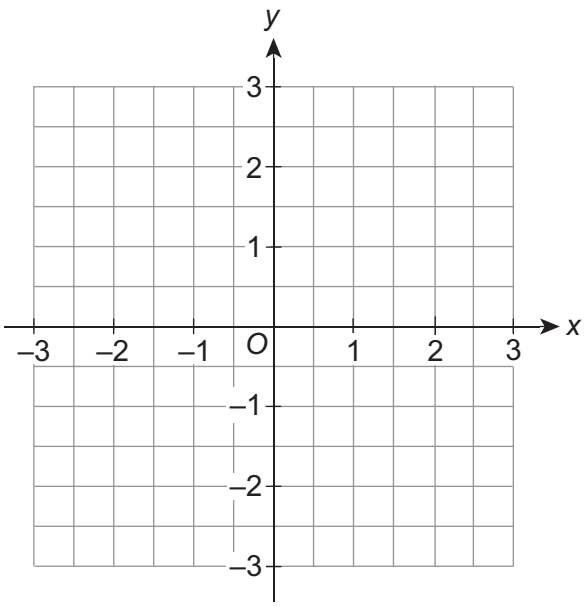
Evaluate $\int_{-1}^{\sqrt{3}-2} \frac{x}{x^2+4x+5}\,dx$. Give your answer in the form $\log_e(a) + b$, where $a, b \in R$.

Question 10 (3 marks)

A particle moves along a straight line. At time t seconds, the particle's position, in centimetres, from the fixed point O is given by $\underset{\sim}{r}(t) = 2\cos(2t)\underset{\sim}{i} + \left(\sin^2(t) - 1\right)\underset{\sim}{j}$, where $t \in \left[0, \frac{\pi}{2}\right]$ and $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are unit vectors of length 1 m.

a. Show that the cartesian equation of the path is $x + 4y = -2$. 1 mark

b. Sketch the path of the particle on the axes below. Label the endpoints and axis intercepts with their coordinates and show the direction of motion. 1 mark



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c. At what time is the speed of the particle first equal to $\frac{\sqrt{17}}{2}$ m s^{-1}? 1 mark

End of examination questions

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